

3.5 The Comparison Test & Limit Comparison Test

Motivation: How could we show the series $\sum_{n=1}^{\infty} \left(\frac{1}{1+2^n} \right)$ to be convergent?

Theorem (The Comparison Test):

Proof: The proof is very similar to what is mentioned in the motivation section.

Note: When we say “for all n ” in the above theorem we implicitly mean all the possible values that n can take on as an index. The theorem still holds if the index n starts at a value different from one.

Example 1: Determine (with justification) whether the series $\sum_{k=1}^{\infty} \left(\frac{\ln(k)}{k} \right)$ converges or diverges.

Example 2 (you try): Determine (with justification) whether the series...

(a) $\sum_{n=1}^{\infty} \left(\frac{1}{2n^2 + n} \right)$

(b) $\sum_{n=1}^{\infty} \left(\frac{\arctan(n)}{n^{1.8}} \right)$ converges or diverges.

Theorem (The Limit Comparison Test):

Proof:

Example 3: Determine (with justification) whether the series $\sum_{n=1}^{\infty} \left(\frac{1}{2^n - 1} \right)$ converges or diverges.

Example 4 (you try): Determine (with justification) whether the series $\sum_{n=1}^{\infty} \left(\frac{1}{2n + 3} \right)$ converges or diverges.

Example 5: Determine (with justification) whether the series $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$ converges or diverges.